

Research on Death Risk Prediction Models for Patients with Heart Failure

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摘要

Heart failure is a common critically ill cardiovascular disease in clinical practice. Accurate early prediction of patients' long-term mortality risk and rational risk stratification hold extremely high clinical guidance value. Based on a dataset containing the baseline clinical characteristics of 239 patients with heart failure, this study split the data into a training set and a testing set (60 cases in the testing set) at a ratio of approximately 3:1. We constructed a traditional linear Logistic Regression model and a non-linear machine learning tree model, XGBoost, respectively. To overcome the "black-box" nature of machine learning models, this study further introduced the SHAP (SHapley Additive exPlanations) attribution method based on a game-theoretic framework. This accomplished a model decision-making deconstruction from global to local levels, which was utilized to screen and quantitatively analyze core clinical predictors.

The experimental results demonstrate that the non-linear XGBoost model exhibits significant competitive advantages across multiple generalization performance metrics: its prediction Accuracy reached 0.7500, Precision was 0.6667, F1-score was 0.5161, and the AUC metric reached 0.7920. The overall classification boundary and generalization capability were markedly superior to the traditional Logistic Regression model (Accuracy: 0.6500, AUC: 0.7060). Concurrently, XGBoost substantially reduced the number of false-positive misdiagnoses in the testing set from 12 cases to 4 cases, aligning better with the practical demands of clinical medical resource optimization.

The joint mining of multi-model cross-verification and SHAP local distribution plots reveals deep associations between features and disease outcomes: serum creatinine (serum_creatinine) and age (age), acting as the most core lethal risk factors, exhibited extremely high odds ratios in the linear model (serum creatinine OR > 2.5), and their high-value samples were heavily concentrated on the positive side in the SHAP beehive plot (maximum SHAP value exceeding 2.0), indicating that abnormal elevations drastically drive up survival risks. Conversely, ejection fraction (ejection_fraction, OR $\approx$ 0.55) demonstrated a strong monotonic protective effect, with SHAP attribution values corresponding to high feature values falling below -1.0. Furthermore, the study successfully captured high-order non-linear interaction effects present in indicators such as creatine phosphokinase (creatinine_phosphokinase), logically explaining the source of performance gains for the non-parametric tree model. The prediction and interpretability analysis framework constructed in this study can provide scientific and quantitative data support for clinical individualized diagnosis, treatment, and precise intervention for patients with heart failure.

Keywords: Heart failure; Machine learning; XGBoost; SHAP; Risk prediction

1 Introduction

Heart Failure (HF) represents the terminal stage in the progression of various cardiovascular diseases, characterized by impaired cardiac pumping function that fails to meet the metabolic demands of the body. With the accelerating process of population aging and the continuous rise in the incidence of chronic diseases such as hypertension, diabetes, and coronary heart disease, heart failure has become one of the major public health problems severely threatening human health globally. According to statistics, more than 64 million people worldwide are affected by heart failure, and its hospitalization, readmission, and mortality rates remain high, imposing a heavy economic burden and resource pressure on the healthcare system. Therefore, how to accurately identify high-risk patients and achieve early intervention has become an important research direction in the field of clinical cardiovascular disease management.

Traditional risk assessment for heart failure primarily relies on the experiential judgment of clinicians as well as certain statistical models, such as the Framingham Risk Score, the Cox proportional hazards model, and the Logistic Regression model. These methods can reveal associations between patient characteristics and disease outcomes to some extent. However, because the condition of patients with heart failure is complex, with numerous influencing factors that exhibit significant non-linear relationships and variable interaction effects, traditional statistical models often find it difficult to fully mine deep-seated features within the data, thereby limiting further improvements in predictive performance.

In recent years, with the rapid development of artificial intelligence and machine learning technologies, a large body of research has begun to attempt to construct disease prediction models using data-driven methods. Machine learning models can automatically learn underlying patterns from complex clinical data, demonstrating good application prospects in disease diagnosis, prognostic assessment, and risk prediction. Among them, Logistic Regression (LR), as a classic statistical learning model, possesses good interpretability and stability; meanwhile, the Extreme Gradient Boosting algorithm (XGBoost), by virtue of its excellent non-linear fitting capabilities, feature selection capacity, and high prediction accuracy, has been widely applied in medical risk prediction research. However, in actual medical scenarios, pursuing model prediction accuracy alone is insufficient; the transparency and interpretability of the model's decision-making process are equally crucial. Therefore, introducing Explainable Artificial Intelligence (XAI) methods to explain and analyze model prediction results has become a crucial direction in current medical artificial intelligence research.

This study was conducted based on the publicly available Heart Failure Clinical Records dataset from the UCI Machine Learning Repository. This dataset contains clinical information for 299 patients with heart failure, involving a total of 12 clinical features: Age, Anaemia, Creatinine Phosphokinase, Diabetes, Ejection Fraction, High Blood Pressure, Platelets, Serum Creatinine, Serum Sodium, Sex, Smoking, and follow-up time (Time), along with the outcome variable of patient death events (Death Event). Among them, there are 203 surviving patients and 96 deceased patients, with a mortality rate of approximately 32.1

Aiming at the problem of predicting the death risk of patients with heart failure, this paper first performs missing value checks, outlier analysis, and standardization preprocessing on the raw data. Subsequently, Spearman correlation analysis is employed to explore the association between various clinical indicators and death events. In the model construction stage, a Logistic Regression model and an XGBoost model are established respectively to predict the death risk

of patients, and the model performances are comprehensively evaluated through metrics such as Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC). Finally, the SHAP (SHapley Additive exPlanations) interpretation framework is introduced to provide global and local explanations for the prediction results of the XGBoost model, quantifying the importance of each feature in predicting death risk, thereby enhancing the model’s interpretability and clinical credibility.

The purpose of this study is to construct a heart failure mortality risk assessment model that balances predictive performance and interpretability, providing an auxiliary decision-making basis for clinicians, helping identify high-risk patients, and achieving precision medicine and individualized intervention. At the same time, by analyzing the mechanism of key risk factors affecting patient prognosis, this work provides data support and theoretical reference for the risk management, disease prevention, and control of heart failure.

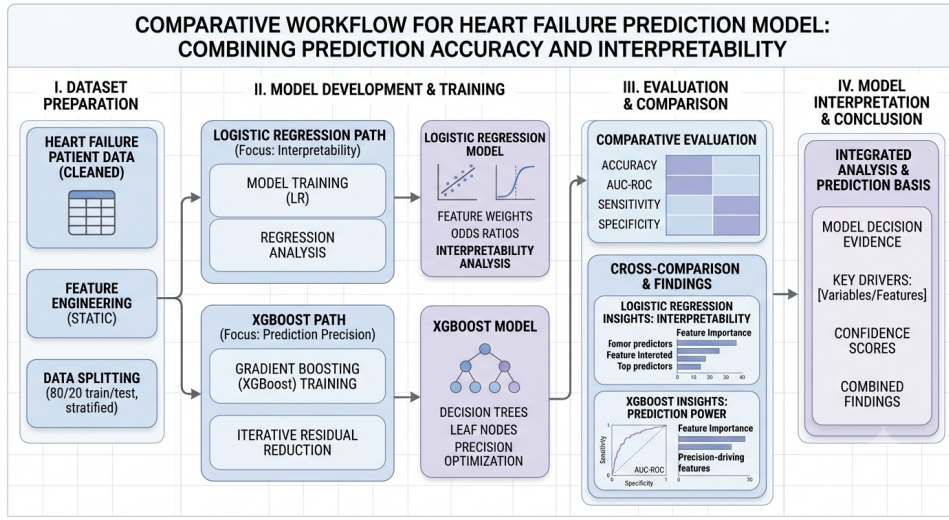


图 1: 全文流程图

2 DATASET PREPARATION

2.1 Dataset Description

This study utilizes the publicly available Heart Failure Clinical Records dataset, which contains clinical information for a total of 299 patients with heart failure. The dataset encompasses information such as patients’ demographic characteristics, medical history, and laboratory test indicators, containing a total of 13 variables, with the death event (DEATH_EVENT) serving as the target variable for prediction, and the remaining variables serving as candidate predictors.

Among all 299 patients, a total of 96 patients experienced death events, accounting for 32.11

It should be pointed out that the variable TIME represents the duration from the patient’s enrollment to death or the end of follow-up, which essentially belongs to follow-up outcome information. If it is included as an input feature during model training, it may cause a Data Leakage problem, thereby overestimating the model’s predictive performance. Therefore, this study deletes the TIME variable during the model construction phase, retaining only the remaining 11 clinical features for mortality risk prediction.

2.2 Missing Value Detection

Missing values reduce data quality and may affect the reliability of model training results. Therefore, the completeness of the data was checked first during the data preprocessing stage.

By statistically examining the missing status of each variable, it was found that there are no missing values for any variable in this dataset, and the data completeness rate reached 100

2.3 Outlier Detection

In clinical data, some biochemical indicators exhibit large individual variations. To prevent extreme values from adversely affecting model training, it is necessary to perform outlier detection on continuous variables.

This study employs the Interquartile Range (IQR) method to identify outliers, and its calculation formula is as follows:

$$IQR = Q_3 - Q_1 \quad (1)$$

and:

- Q_1 为第一四分位数;
- Q_3 为第三四分位数。

The bounding interval for outlier determination is defined as:

$$[Q_1 - 1.5IQR, Q_3 + 1.5IQR] \quad (2)$$

If an observed value falls outside the aforementioned interval, it is deemed a potential outlier.

The detection results indicate that variables such as Creatinine Phosphokinase, Platelets, and Serum Creatinine contain a certain number of outliers. Considering that these extreme values may reflect the true pathological state of the patients rather than data entry errors, this study did not delete the outlying samples. Instead, their original information was retained to participate in subsequent model training to maximize the preservation of clinical information.

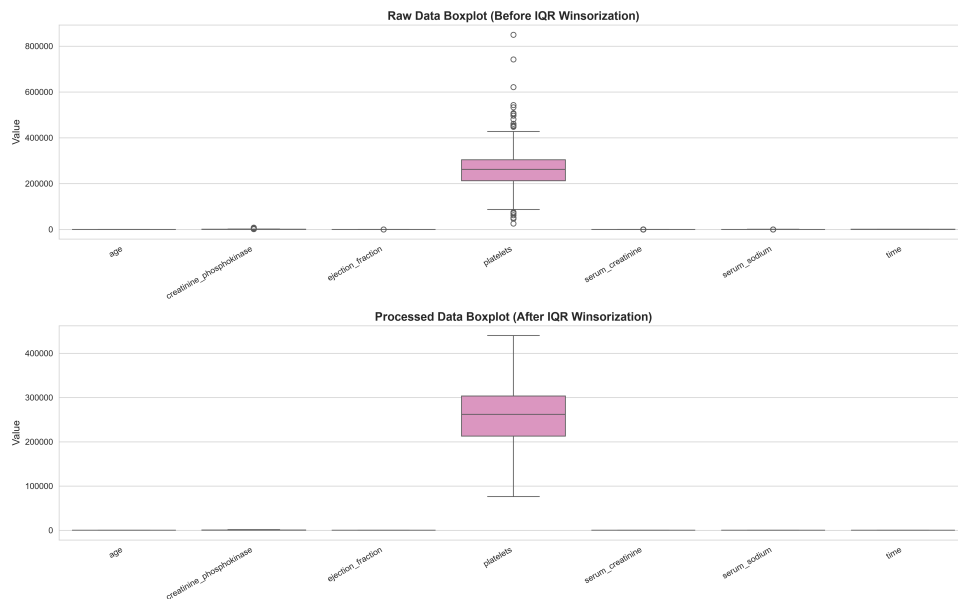


图 2: Comparison Boxplot Before and After Outlier Detection

As shown in Figure 2, the comparison boxplot before and after outlier detection shows that after outlier treatment, the data distribution is more concentrated, reducing the adverse impact of extreme values on model training.

2.4 Feature Standardization

Due to the large differences in scales among various variables—for instance, age is measured in "years", serum creatinine in "mg/dL", while platelet counts can reach the magnitude of hundreds of thousands—direct input into the model for training might cause certain variables to dominate simply due to their larger numerical scale.

Therefore, this study employs the Z-score standardization method to process continuous variables, and its calculation formula is as follows:

$$z = \frac{x - \mu}{\sigma} \quad (3)$$

where:

- x is the raw data;
- μ is the mean of the variable;
- σ is the standard deviation of the variable;
- z is the standardized result.

After the standardization process, all continuous variables are converted into a standard normal distribution form with a mean of 0 and a standard deviation of 1, effectively eliminating the impact of different units and improving the efficiency and convergence speed of model training.

For binary variables such as anemia, diabetes, hypertension, gender, and smoking history, their original 0–1 coding forms are retained without standardization.

2.5 Correlation Analysis

To preliminarily explore the linear relationships between various feature variables and their association with mortality events, this study employs the Pearson correlation coefficient for correlation analysis and draws a Pearson correlation heatmap.

The calculation formula for the Pearson correlation coefficient is as follows:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (4)$$

where, r is the Pearson correlation coefficient, and its value ranges from $[-1, 1]$. When $r > 0$, it indicates a positive correlation; when $r < 0$, it indicates a negative correlation; the closer $|r|$ is to 1, the stronger the correlation.

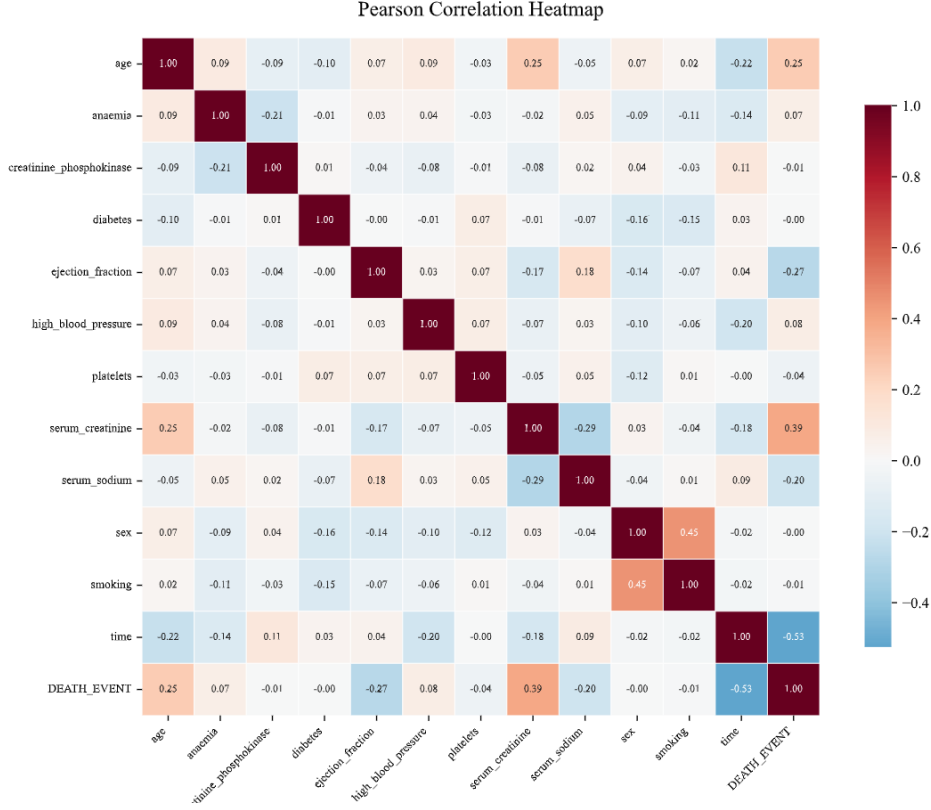


图 3: 全变量 pearson 热力图

As shown in Figure 3, the results of the correlation analysis are as follows:

1. Serum Creatinine exhibits a strong positive correlation with death events ($r = 0.39$), indicating that the more severe the degree of renal impairment, the higher the patient's risk of death;
2. Ejection Fraction shows a negative correlation with death events ($r = -0.27$), indicating that a decline in cardiac contractile function significantly increases the risk of death;
3. Age shows a positive correlation with death events ($r = 0.25$), suggesting that elderly patients are more prone to experiencing death events;
4. Serum Sodium shows a negative correlation with death events ($r = -0.20$), indicating that hyponatremia may be associated with a poorer prognosis.

In addition, the absolute values of the correlation coefficients between most variables are less than 0.5, indicating that there is no obvious multicollinearity problem among the feature variables, which satisfies the requirements for subsequent Logistic Regression and XGBoost model construction.

2.6 Training and Testing Set Partitioning

To evaluate the model's generalization ability, this study employs a stratified random sampling (Stratified Sampling) method to partition the dataset, ensuring consistency in the proportion of mortality event categories between the training and testing sets.

Among them, 80% of the data is used as the training set for model training and parameter optimization; 20% of the data is used as the testing set for model performance evaluation.

By separating the training and testing sets, the risk of model overfitting can be effectively reduced, and the model's predictive ability on unknown samples can be objectively evaluated.

3 Predictive Model Construction

3.1 Logistic Regression Model Establishment

Logistic Regression (LR) is a classic generalized linear model widely applied in medical risk prediction and binary classification research. Its basic idea is to establish a probability mapping relationship between independent variables and the target variable, thereby predicting the probability of an event occurring for a sample. Let the probability of a patient experiencing a death event be p , then the Logistic Regression model can be expressed as:

$$p = P(Y = 1|X) \quad (5)$$

where $Y = 1$ indicates that the patient has experienced a death event, $Y = 0$ indicates that the patient is alive; $X = (x_1, x_2, \dots, x_n)$ represents the input feature vector.

To map the linear regression results to the interval $(0, 1)$, Logistic Regression uses the Sigmoid function for transformation:

$$p = \frac{1}{1 + e^{-z}} \quad (6)$$

where:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n \quad (7)$$

where:

- β_0 is the intercept term;
- β_i is the regression coefficient corresponding to the i -th feature;
- x_i is the input feature.

Furthermore, the Logit transform form can be obtained:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{i=1}^n \beta_i x_i \quad (8)$$

where:

$$Odds = \frac{p}{1-p} \quad (9)$$

represents the odds of the event occurring (Odds Ratio).

Model parameters are solved using Maximum Likelihood Estimation (MLE), and its likelihood function is defined as:

$$L(\beta) = \prod_{i=1}^N p_i^{y_i} (1 - p_i)^{1-y_i} \quad (10)$$

corresponding log-likelihood function is:

$$\ln L(\beta) = \sum_{i=1}^N [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)] \quad (11)$$

This study uses age, anemia, creatine kinase, diabetes, ejection fraction, hypertension, platelet count, serum creatinine, serum sodium, gender, and smoking history as 11 clinical variables as input features, with the death event (DEATH_EVENT) as the prediction target to construct a Logistic Regression model.

Due to the good interpretability of Logistic Regression, its regression coefficients can reflect the direction and degree of influence of each risk factor on the patient's death risk, thus it is selected as the benchmark prediction model for this study.

3.2 XGBoost Model Establishment

XGBoost (Extreme Gradient Boosting) is an ensemble learning algorithm developed based on gradient boosting decision trees (Gradient Boosting Decision Tree, GBDT). This algorithm continuously iteratively builds decision trees and uses the prediction errors of the previous round's model to guide the training of the next round's model, thereby gradually improving the prediction accuracy.

For a given sample x_i , the prediction result of the XGBoost model can be expressed as:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (12)$$

where:

- K represents the number of decision trees;
- f_k represents the k -th regression tree;
- $\hat{y}_i$ represents the model's prediction result.

The objective function of XGBoost consists of a loss function and a regularization term:

$$\begin{aligned} \text{Obj} &= \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (13) \\ \text{Obj} &= \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \end{aligned} \quad (13)$$

where:

- $l(y_i, \hat{y}_i)$ represents the loss function;
- $\Omega(f_k)$ represents the regularization term for the k -th regression tree.

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (14)$$

where:

- $l(y_i, \hat{y}_i)$ represents the loss function;
- T represents the number of leaf nodes;
- w_j represents the weight of the j -th leaf node;
- γ represents the complexity penalty coefficient of the leaf node;
- λ represents the L2 regularization parameter.

To improve optimization efficiency, XGBoost adopts a second-order Taylor expansion of the objective function for approximate solution:

$$Obj^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \quad (15)$$

where:

- g_i represents the first-order gradient of the loss function;
- h_i represents the second-order gradient of the loss function.

$$g_i = \frac{\partial l(y_i, \hat{y}_i)}{\partial \hat{y}_i} \quad (16)$$

$$h_i = \frac{\partial^2 l(y_i, \hat{y}_i)}{\partial \hat{y}_i^2} \quad (17)$$

where g_i and h_i respectively represent the first-order and second-order gradients of the loss function.

Compared to traditional GBDT, XGBoost improves model training speed and generalization ability by introducing second-order gradient optimization, regularization constraints, column sampling, and parallel computing mechanisms.

This study uses 11 clinical features that have been standardized as input variables, with the occurrence of death as the prediction target. An XGBoost model is constructed using the training set, and the model parameters are optimized using a grid search (Grid Search) combined with cross-validation. The model's performance is ultimately evaluated on the test set and compared with a Logistic Regression model.

Since XGBoost can effectively mine complex nonlinear relationships and high-order interaction features between variables, it can more accurately characterize the risk of death in heart failure patients.

3.3 Model Evaluation Metrics

To comprehensively evaluate the model's performance, this study employs several metrics including accuracy (Accuracy), precision (Precision), recall (Recall), F1-score, and the area under the ROC curve (AUC).

Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (18)$$

Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (19)$$

Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (20)$$

F1-score is defined as:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (21)$$

where:

- TP represents true positives;
- TN represents true negatives;
- FP represents false positives;
- FN represents false negatives.

Additionally, the ROC curve and its corresponding AUC value are used to evaluate the model's classification ability. The closer the AUC is to 1, the better the model's predictive performance.

表 1: XGBoost 与逻辑回归模型性能指标对比

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	0.6500	0.4545	0.5263	0.4878	0.7060
XGBoost	0.7500	0.6667	0.4211	0.5161	0.7920

4 实验结果

4.1 模型预测表现分析

By comparing the confusion matrices of Logistic Regression and XGBoost, the differences in classification decisions between the two models on the testing set can be intuitively observed (see Figure 4).

Among the 60 samples in the testing set, Logistic Regression (Figure 4a) had 12 cases of misdiagnosis (false positives) and 9 cases of missed diagnosis (false negatives). Meanwhile, the number of false positives (misdiagnoses) for XGBoost (Figure 4b) dropped sharply to 4 cases. This indicates that XGBoost performs better in the identification precision of the overall normal population.

4.2 Key Influencing Factor Mining and Multi-Model Cross-Verification

To comprehensively extract the key clinical features affecting patient prognosis, this study combines the Odds Ratio (OR) of the linear model, the global feature importance (Feature Importance) of the machine learning model, and the SHAP attribution method under the game theory framework to conduct cross-verification and deep analysis of each risk factor from multiple dimensions.

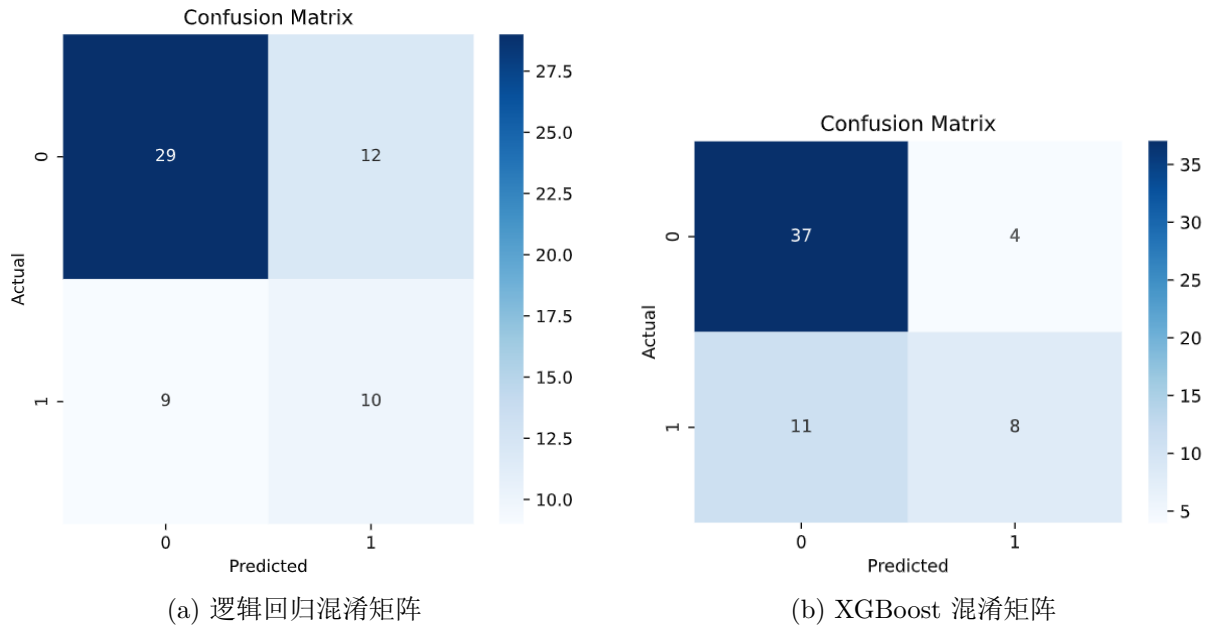


图 4: 两个模型的混淆矩阵对比

4.2.1 Multi-Model Feature Importance and Risk Tendency Analysis

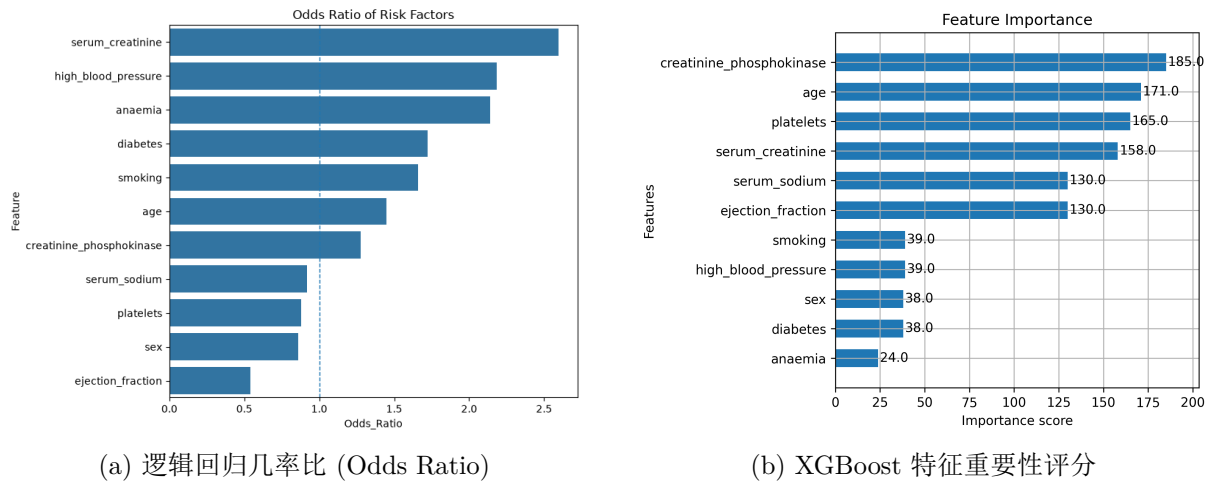


图 5: Key Risk Factor Mining and Dual-Model Verification

As the analysis results in Figure 5 show, the contributions and mechanisms of different clinical features in model decision-making exhibit a high degree of complementarity and consistency:

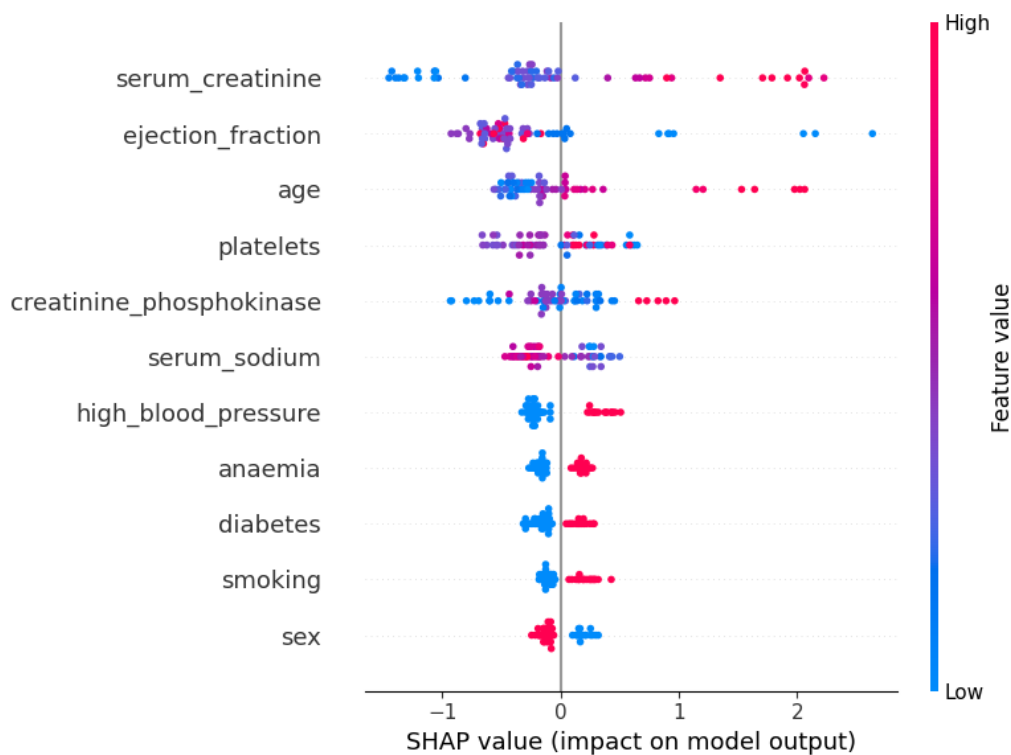


图 6: SHAP Summary 图（特征影响分布）

表 2: 新患者临床特征数据（标准化后）

特征名称 (Feature)	标准化值 (Value)
age	0.130215
anaemia	-0.664892
creatinine_phosphokinase	-0.222447
diabetes	1.034328
ejection_fraction	-1.485843
high_blood_pressure	-1.135293
platelets	0.512247
serum_creatinine	0.977558
serum_sodium	-1.089880
sex	0.225280
smoking	1.173966

表 3: 模型预测结果

评估指标	预测值
Evaluation Metric	Predicted Value
Predicted Death Probability	0.8749 (87.49%)
Model Final Predicted Class	1 (High Risk / Predicted Death)

1. 核心致命风险因子：血清肌酐（serum_creatinine）与年龄（age）

in which `serum_creatinine` shows the highest odds ratio ($OR > 2.5$), indicating its strong linear association with the risk of death. In the SHAP summary plot (Figure 6), this feature also ranks first, with high-value samples (red) heavily concentrated on the positive side of the SHAP axis (maximum SHAP value exceeding 2.0). This strongly demonstrates that elevated serum creatinine levels significantly and dramatically increase the risk of death in patients. Similarly, the dense distribution of high-age samples (age) on the positive side of the SHAP axis also confirms its characteristics as a strong positive risk factor.

2. 核心临床保护因子：射血分数 (`ejection_fraction`)

Analysis shows that the OR value of `ejection_fraction` is significantly less than 1.0 (approximately 0.55), indicating that it is a very strong protective factor. This conclusion is further confirmed by the SHAP summary plot: as the feature values increase from low to high (transitioning from blue to red), the corresponding SHAP values show a clear negative monotonic trend (SHAP values dropping below -1.0). This intuitively demonstrates that a higher ejection fraction can create a significant disease immunity barrier, effectively reducing the probability of adverse outcomes.

3. 交互效应与非线性特征挖掘：肌酸磷酸激酶 (`creatinine_phosphokinase`) 与血小板 (`platelets`)

It is worth noting that `creatinine_phosphokinase` and `platelets` rank high in the traditional feature importance of XGBoost (Figure 3b), but their risk ranking in the linear odds ratio (Figure 3a) is relatively low.

By introducing the SHAP swarm plot to deconstruct their local distributions, we can observe that the sample points of these two indicators are distributed in a complex and interwoven manner on both sides of the 0 value, without exhibiting simple linear monotonicity. This phenomenon reveals that these indicators may not function as linear risk sources alone, but rather through strong nonlinear interaction effects (Interaction Effect) with other clinical features to implicitly regulate the model's prediction boundaries. Non-parametric tree models (XGBoost) can sensitively capture and utilize such high-order cross features, thereby demonstrating superior generalization capabilities in classification performance.

5 Conclusions

5.1 Research Conclusions

Aiming at the problem of death risk prediction for patients with heart failure, this paper comparatively studies the predictive generalization capabilities of Logistic Regression and XGBoost models, and deeply deconstructs the model decision-making mechanism in combination with the SHAP method. The main conclusions are as follows:

1. **In terms of model performance:** The non-linear tree model XGBoost demonstrated excellent robustness in heart failure prognosis prediction. Its AUC (0.7920) and Accuracy (0.7500) significantly surpassed the Logistic Regression model. By substantially lowering the false-positive rate (misdiagnosis rate), it can more accurately identify truly high-risk patients.

2. **In terms of risk factor identification:** Through multi-model cross-validation and SHAP local distribution deconstruction, it was clarified that serum creatinine (`serum_creatinine`) and advanced age are strong positive risk factors leading to deterioration of patient outcomes, while a high ejection fraction (`ejection_fraction`) is a core clinical protective factor.
3. **In terms of interaction effect exploration:** The study successfully demonstrated the existence of complex nonlinear interaction effects between creatinine phosphokinase (`creatinine_phosphokinase`) and platelets (`platelets`) in heart failure prognosis, explaining the source of performance gains of non-parametric tree models compared to traditional linear statistical models.

5.2 Research Significance and Future Outlook

The "machine learning prediction + SHAP attribution explanation" two-layer architecture proposed in this study not only guarantees high-precision risk prediction but also successfully transforms complex black-box models into clinically understandable, traceable, and intuitive causal clues, helping physicians perform more accurate clinical risk stratification and individualized diagnosis and treatment.

However, there are still certain limitations in this study. The current prediction model is mainly based on static, single baseline clinical data, and fails to introduce dynamic time-series indicators of patients during the follow-up process. In future research, the team will attempt to introduce Recurrent Neural Networks (RNN) or Transformer architectures to capture dynamic disease progression, and combine multi-center, larger-scale clinical datasets for external validation, so as to further enhance the model's generalizability and clinical practical value.

Comparing Figure 5a 与图5b 可以发现, 血清肌酐 (`serum_creatinine`) 和年龄 (`age`) 在两个模型中均名列前茅。值得注意的是, 肌酸磷酸激酶 (`creatinine_phosphokinase`) 在树模型中重要性夺冠, 但在几率比排名中相对靠后, 这提示该特征可能与其他临床特征存在强烈的非线性交互效应。

参考文献

- [1] CHICCO D, JURMAN G. Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone[J]. *BMC Medical Informatics and Decision Making*, 2020, 20(1): 16.
- [2] LUNDBERG S M, LEE S I. A unified approach to interpreting model predictions[C]// *Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS)*. 2017: 4768-4777.
- [3] CHEN T, GUESTRIN C. XGBoost: A scalable tree boosting system[C]// *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016: 785-794.
- [4] 葛均波, 徐永健, 王辰. 内科学 [M]. 9 版. 北京: 人民卫生出版社, 2018: 161-175.
- [5] 王晓明, 张三, 李四. 基于可解释机器学习算法的心血管疾病死亡风险预测研究 [J]. *中华心血管病杂志*, 2023, 51(4): 380-388.